

Project Design Phase

Solution Architecture

Date	28 September 2026
Team ID	SWTID-2026-5454
Project Name	Streamlining IT Procurement: Automating Standard Laptop Orders with Flow Designer
Maximum Marks	4 Marks

Solution Architecture:

The solution architecture for this project bridges the IT procurement problem with a ServiceNow-based automation solution. It defines how the Standard Laptop request moves through the Service Catalog, approval process, Flow Designer, automated Catalog Task creation, and Hardware team assignment.

Architecture and Data Flow

Employee / IT User		ServiceNow Service Catalog		Approver
Submit Standard Laptop Request		Requested Item		Approve Request
		↓		
		Flow Designer		
		Create Catalog Task Short Description: Laptop need to Configured Assignment Group: Hardware		Hardware Team

Figure 1: Solution architecture and data flow for Standard Laptop order automation

Architecture Components

Component	Technology / Service	Purpose
User Interface	ServiceNow Service Catalog	Allows the employee / IT user to submit the Standard Laptop request and view request information.
Request & Approval	Service Catalog + Approval Process	Captures the request and obtains approval before fulfillment task creation.
Automation Logic	ServiceNow Flow Designer	Triggers the automated fulfillment workflow after the approved request.
Task Creation	Create Catalog Task action	Creates the Catalog Task for the Standard Laptop configuration.
Task Assignment	Assignment Group: Hardware	Routes the generated configuration task to the Hardware team.
Data / Records	ServiceNow platform data tables	Stores the requested item and catalog task information used by the workflow.

Process Flow

1. Employee / IT user selects Standard Laptop from the Service Catalog and submits the request.
2. The requested item enters the approval process.
3. After approval, the Standard Laptop Task Flow is triggered through the catalog item's Process Engine.
4. Flow Designer uses the Create Catalog Task action to create the configuration task.
5. The task is given the description "Laptop need to Configured" and assigned to the Hardware group.

6. The Hardware team receives the task and performs the required laptop configuration.
7. The Requested Item and Catalog Tasks sections are used to verify the request and task status.

Architecture Characteristics

Characteristic	Project Implementation
Automation	Flow Designer automatically creates the fulfillment Catalog Task after approval.
Integration	Service Catalog, approval process, Flow Designer, and Hardware assignment work as one workflow.
Reliability	The defined flow provides a consistent task-creation and assignment process.
Maintainability	The flow and catalog-item process configuration can be updated within ServiceNow.
Scalability	The same workflow pattern can be extended to other IT procurement catalog items and fulfillment tasks.

Conclusion

The proposed architecture provides a clear connection between the Standard Laptop procurement requirement and the ServiceNow technology solution. By combining Service Catalog, approval processing, Flow Designer, automatic Catalog Task creation, and Hardware assignment, the architecture supports timely configuration and reduces manual intervention.